

Chapter-5

System Modeling

Topics covered

- ⑩ 1. Context models
- ⑩ 2. Interaction models
- ⑩ 3. Structural models
- ⑩ 4. Behavioral models
- ⑩ 5. Model-driven engineering

Introductory Slides: System modeling

- ⑩ System modeling is the process of developing abstract models of a system, with each model presenting a different view or perspective of that system.
- ⑩ System modeling has now come to mean representing a system using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML).
- ⑩ System modeling helps the analyst to understand the functionality of the system and models are used to communicate with customers.

Existing and Planned System Models

- ⑩ Models of the existing system are used during requirements engineering.
 - They help clarify what the existing system does and can be used as a basis for discussing its strengths and weaknesses.
 - These then lead to requirements for the new system.
- ⑩ Models of the new system are used during requirements engineering to help explain the proposed requirements to other system stakeholders.
 - Engineers use these models to discuss design proposals and to document the system for implementation.
- ⑩ In a model-driven engineering process, it is possible to generate a complete or partial system implementation from the system model.

System Perspectives

- ⑩ An **external perspective**, where you model the context or environment of the system.
- ⑩ An **interaction perspective**, where you model the interactions between a system and its environment, Or between the components of a system.
- ⑩ A **structural perspective**, where you model the organization of a system or the structure of the data that is processed by the system.
- ⑩ A **behavioral perspective**, where you model the dynamic behavior of the system and how it responds to events.

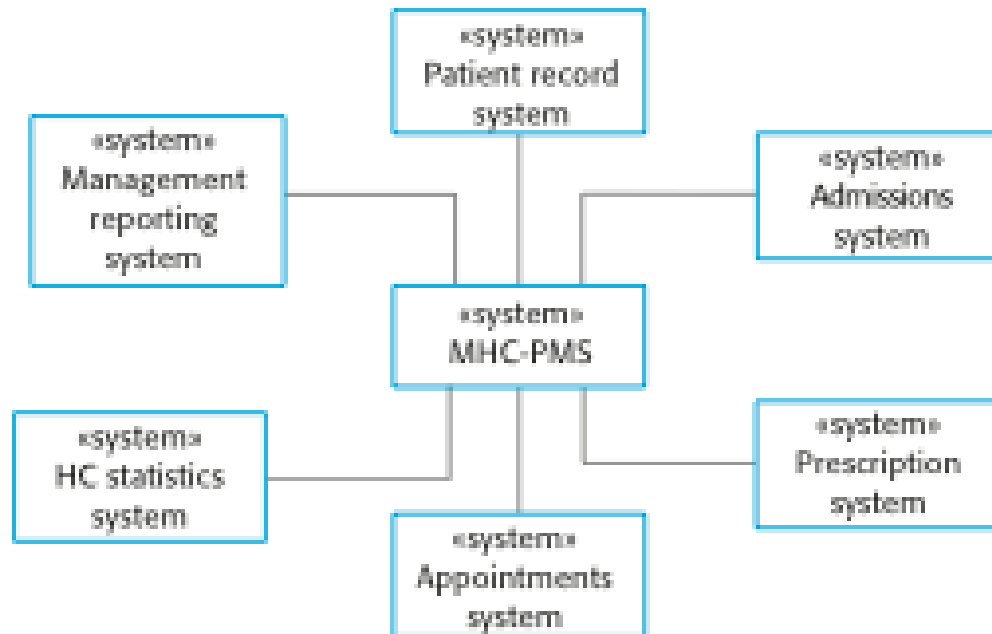
UML Diagram Types

- ⑩ **Activity diagrams**, which show the activities involved in a process or in data processing .
- ⑩ **Use case diagrams**, which show the interactions between a system and its environment.
- ⑩ **Sequence diagrams**, which show interactions between actors and the system and between system components.
- ⑩ **Class diagrams**, which show the object classes in the system and the associations between these classes.

1. Context Models

- ⑩ Context models are used to illustrate the operational context of a system -
- ⑩ They show what lies outside the system boundaries.
- ⑩ Social and organizational concerns may affect the decision on where to position system boundaries.
- ⑩ Architectural models show the system and its relationship with other systems.

The Context of the MHC-PMS



Process Perspective

- ⑩ Context models simply show the other systems in the environment, not how the system being developed is used in that environment.
- ⑩ Process models reveal how the system being developed is used in broader business processes.
- ⑩ How it 'works' Detailed.
- ⑩ UML activity diagrams may be used to define business process models.

2. Interaction Models

- ⑩ Modeling user interaction is used to identify user requirements.
- ⑩ We see structural connections and dynamic (behavioral) interactions.
- ⑩ We do this with graphical models.
- ⑩ Use case diagrams and sequence diagrams may be used for interaction modeling.
- ⑩ These are the most popular modeling mechanisms

Use Case Modeling (Interaction Model)

- ⑩ Use cases were developed originally to support requirements elicitation and now incorporated into the UML.
- ⑩ Each use case represents a discrete task that involves external interaction with a system.
- ⑩ Actors in a use case may be people, devices, or other systems.
- ⑩ Represented diagrammatically to provide an overview of the use case and in a more detailed textual form.

Transfer-data Use Case Diagram (graphical model)

10 A use case in the MHC-PMS

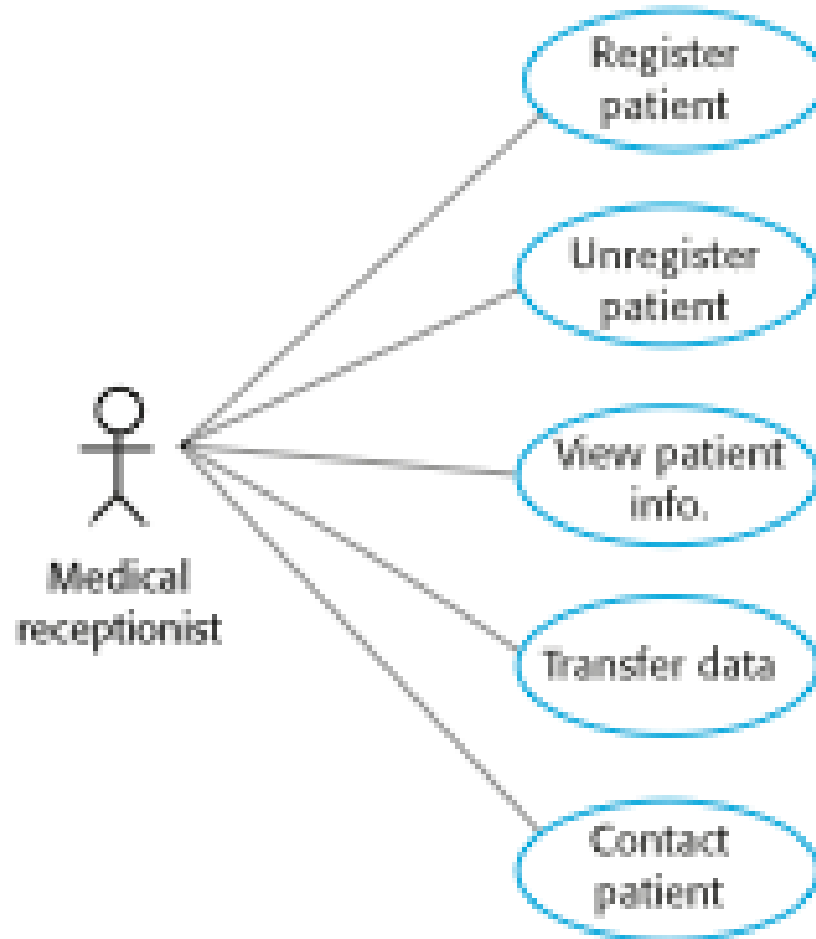


Tabular Description of the 'Transfer data' use-case

MHC-PMS: Transfer data

Actors	Medical receptionist, patient records system (PRS)
Description	A receptionist may transfer data from the MHC-PMS to a general patient record database that is maintained by a health authority. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment.
Data	Patient's personal information, treatment summary
Stimulus	User command issued by medical receptionist
Response	Confirmation that PRS has been updated
Comments	The receptionist must have appropriate security permissions to access the patient information and the PRS.

Use Cases in the MHC-PMS involving the role 'Medical Receptionist' (only showing one actor here)

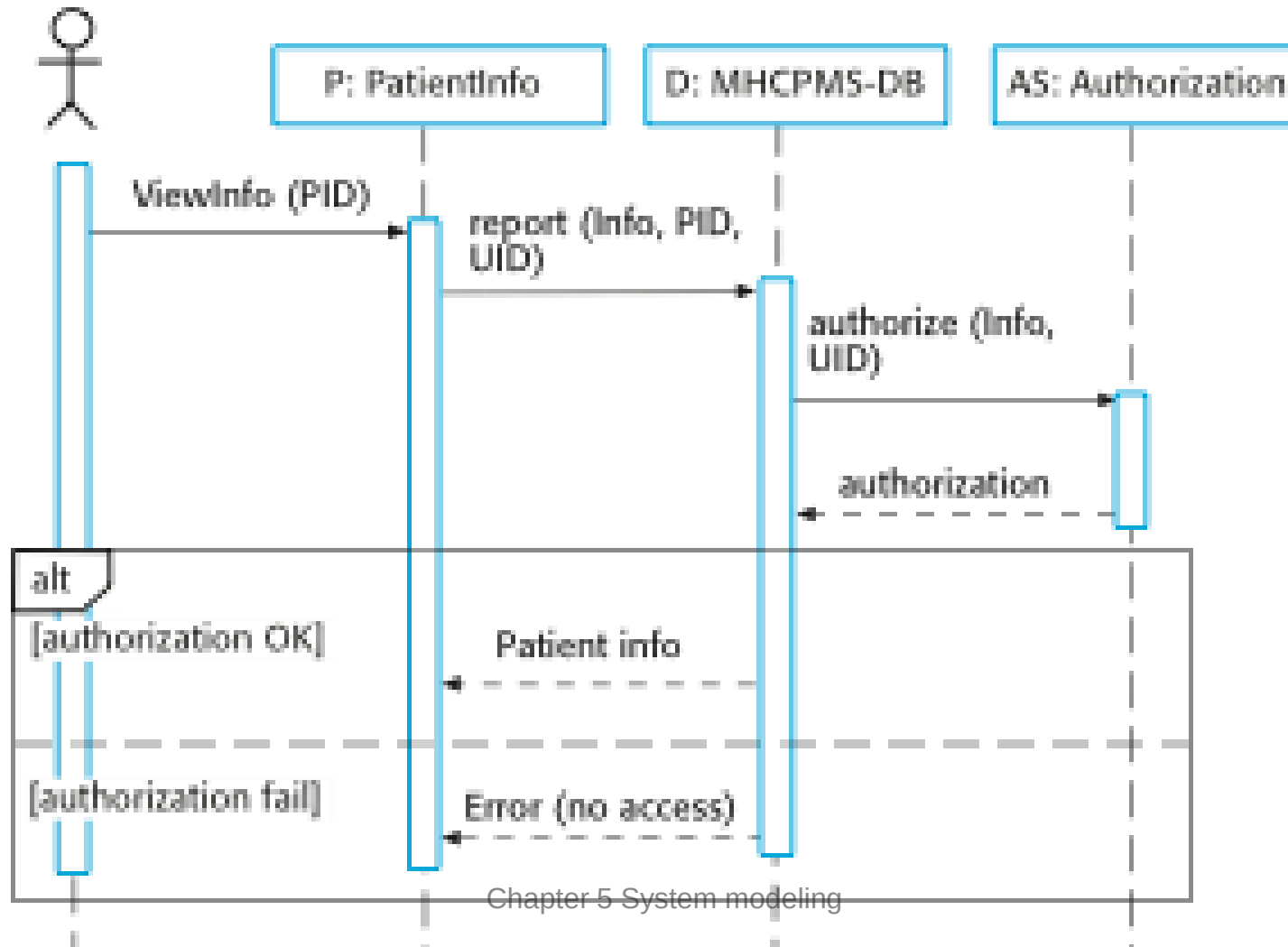


Sequence Diagrams (Interaction Model)

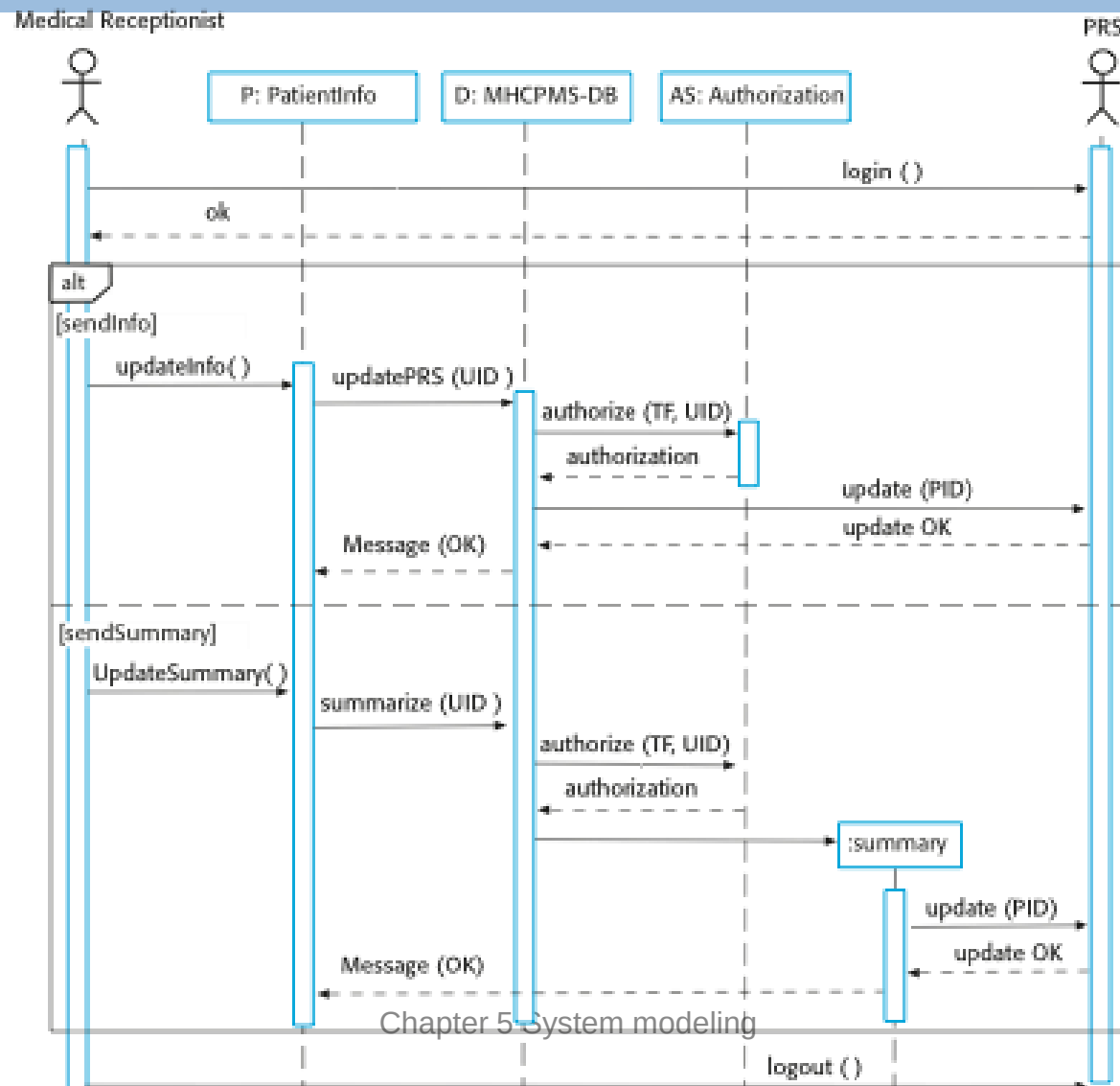
- ⑩ Sequence diagrams are part of the UML and are used to model the interactions between the actors and the objects within a system.
- ⑩ A sequence diagram shows the sequence of interactions that take place during a particular use case or use case instance.
- ⑩ The objects and actors involved are listed along the top of the diagram, with a dotted line drawn vertically from these.
- ⑩ Interactions between objects are indicated by annotated arrows.

Sequence diagram for View Patient Information

Medical Receptionist

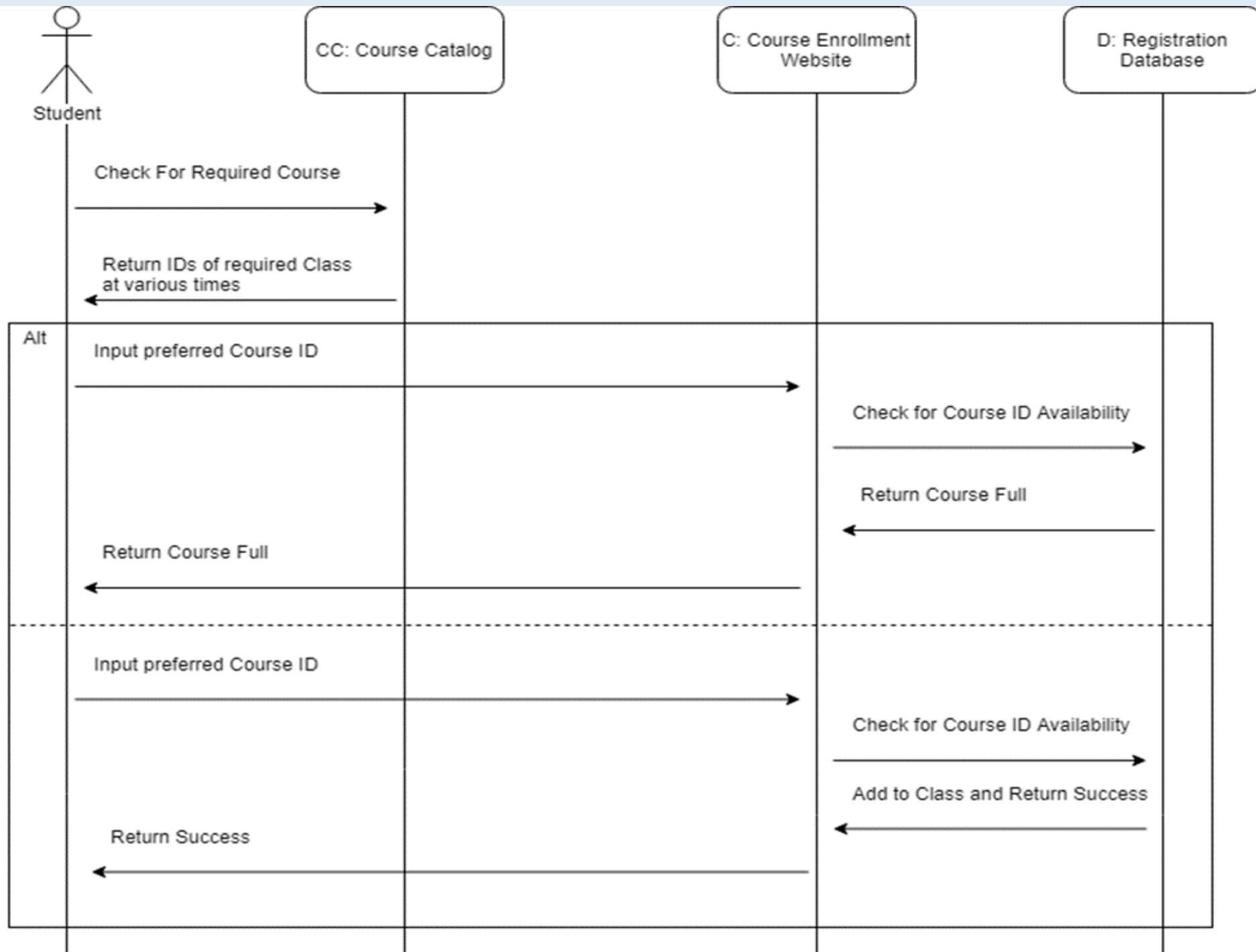


Sequence diagram for Transfer Data



Sequence Diagram

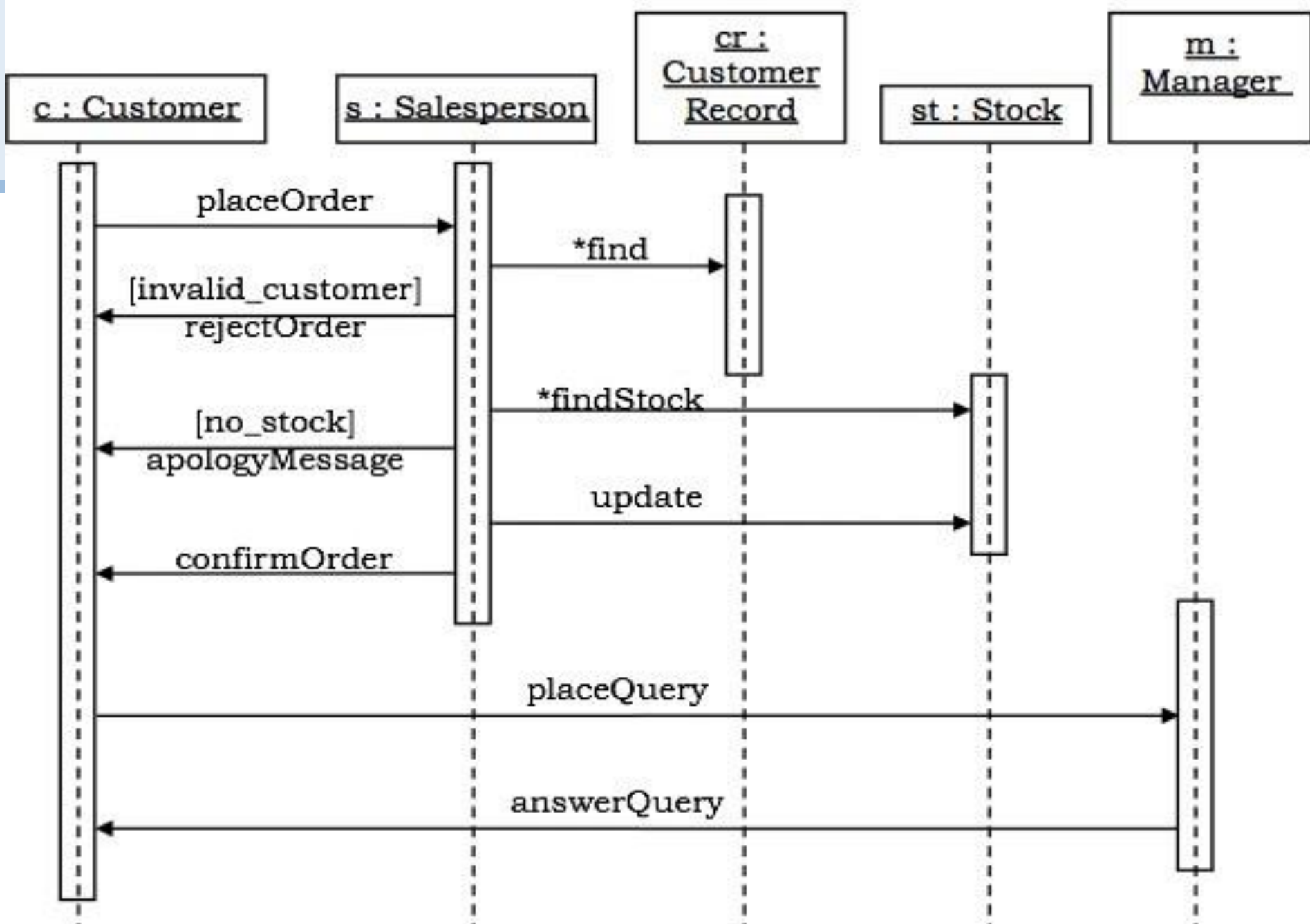
- a sequence diagram showing the interactions involved when a student registers for a course in a university. Courses may have limited enrollment, so the registration process must include checks that places are available. Assume that the student accesses an electronic course catalog to find out about available courses.



Sequence Diagram

An Automated Trading House System:-

- The trading house has transactions with two types of customers, individual customers and corporate customers.
- Once the customer places an order, it is processed by the sales department and the customer is given the bill.
- The system allows the manager to manage customer accounts and answer any queries posted by the customer.



Key points

- ⑩ A model is an abstract view of a system that ignores system details. Complementary system models can be developed to show the system's context, interactions, structure and behavior.
- ⑩ Context models show how a system that is being modeled is positioned in an environment with other systems and processes.
- ⑩ Use case diagrams and sequence diagrams are used to describe the interactions between users and systems in the system being designed. Use cases describe interactions between a system and external actors; sequence diagrams add more information to these by showing interactions between system objects.